

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2013–14

Paper: CSB-604C — Operations Research

(Major Elective Course – I)

Time: Three Hours

Maximum Marks: 70

*Instructions: Attempt **five** questions including Question 1 which is compulsory. Use of calculator is allowed. All undefined symbols have their usual meanings.*

Question 1. Answer any **four** of the following: [4 × 4 = 16]

- (a) Find the graphical solution of the following LPP:

$$\begin{aligned} &\text{Maximize } Z = 2x_1 + 2x_2 \\ &\text{Subject to: } x_1 + x_2 \leq 7 \\ &\quad 0.5x_1 + x_2 \leq 2 \\ &\quad x_1, x_2 \geq 0 \end{aligned}$$

Is it a well-defined LPP?

- (b) How do you calculate the mean and variance of the earliest occurrence time T_E of event i in a project arrow diagram, given a_i , b_i , and m_i as the optimistic, pessimistic, and most likely times respectively?
- (c) What are slack, surplus, and artificial variables? Explain their use for finding the initial basic feasible solution of a LPP.
- (d) What is a non-linear programming problem? Define local and global minima and maxima of a single-variable non-linear function $f(x)$.
- (e) Define the following with an example of each:
- (i) A directed and an undirected cycle
 - (ii) A connected network
 - (iii) An undirected path and an undirected network

Question 2. [6 + 8 = 14]

- (a) Explain the steps of the Simplex method for solving a LPP.
- (b) Solve the following LPP using the Simplex method:

$$\begin{aligned} &\text{Maximize } Z = 3x_1 + 5x_2 + x_3 \\ &\text{Subject to: } 2x_1 + 3x_2 \leq 3 \\ &\quad x_2 + x_3 \leq 8 \\ &\quad 3x_1 + 2x_2 + 4x_3 \leq 15 \\ &\quad x_1, x_2, x_3 \geq 0 \end{aligned}$$

Question 3. If D_{ij} is the duration of an activity (i, j) , write the conditions when activity (i, j) lies on a critical path. Define total float and free float of activity (i, j) . For the following project network, compute the total float, free float, and identify the critical path: $[6 + 8 = 14]$

Activity	Duration (days)	Activity	Duration (days)
1-2	3	3-5	4
1-3	2	4-6	5
2-4	4	5-6	7
2-5	6	5-7	3
3-4	5	6-7	2

What are the uses of total and free floats?

Question 4.

$[6 + 8 = 14]$

- Give the algorithm for processing N jobs through 3 machines, and state the assumptions which must be satisfied for the algorithm to be valid.
- Determine the optimal sequence of 5 jobs on 3 machines A, B, C with no passing between machines. Processing times (in hours) are:

Machine	Job 1	Job 2	Job 3	Job 4	Job 5
A	4	9	8	6	5
B	5	6	2	3	4
C	8	10	6	7	5

Question 5.

$[6 + 4 + 4 = 14]$

- What do you understand by an equipment replacement problem? Name the replacement policies for equipment that fails completely.
- Explain how to find the dual of the general form of a primal LPP.
- With the help of a table, explain how the optimal solutions of a general primal and its dual are related.

Question 6.

$[4 + 10 = 14]$

- What do you understand by “network analysis”? List the problems that can be addressed by network analysis.
- The following network gives links and associated lengths of roads in Seervada Park. Location O is the entrance and T is the end station. Park management wants to install telephone lines to establish phone connections among every pair of stations from O to T, with minimal total miles of line.

Link	Nodes	Length (miles)
1	O – A	2
2	O – B	4
3	O – C	3
4	A – B	1
5	A – D	4
6	B – C	6
7	B – D	2
8	C – E	7
9	D – E	3
10	D – T	4
11	E – T	5

Can this problem be solved by the Minimum Spanning Tree Problem (MSTP) technique?
Give the algorithm of MSTP and find a solution if possible.

Question 7.

$$[4 + 3 + 7 = 14]$$

- (a) What is a convex programming problem?
- (b) What is the direct substitution method of solving a non-linear programming problem?
- (c) Find the optimal solution of the following problem using the method of Lagrangean multipliers:

$$\begin{aligned} \text{Minimize } Z &= 2x_1^2 + x_2^2 + x_3^2 - 4x_1 - 2x_2 - 8x_3 \\ \text{Subject to: } &x_1 + x_2 + x_3 = 2, \quad x_1, x_2, x_3 \geq 0 \end{aligned}$$